0.1 General Linear Group

Definition 0.1.1. Define a *General Linear Group* over a field F as:

$$\mathrm{GL}_n(F) \overset{\text{def}}{=} \{A \in \mathcal{M}_{n\times n}(F) \mid \det A \neq 0\}$$

And, a *Special Linear Group* as:

$$\mathrm{SL}_n(F) \overset{\text{def}}{=} \{A \in \mathcal{M}_{n\times n}(F) \mid \det A = 1\}$$

The property of determinant allows above set are Group under the matrix multiplication.

Lemma 0.1.2. A map

$$\varphi : \mathrm{GL}_n(F) \to F^{\times} : A \mapsto \det A$$

is an Onto Homomorphism with $\ker\varphi = \mathrm{SL}_n(F)$. Therefore,

$$\mathrm{GL}_n(F)/\ker\varphi = \mathrm{GL}_n(F)/\,\mathrm{SL}_n(F) \cong F^{\times}$$

Proof. Well-Defined is obtained by the determinant was obtained by operation in the given field.
And, surjectiveness is given by $\det \mathrm{diag}(1, \ldots, 1, a) = a$ for all $a \in F$.
And homomorphism is obtained by the determinant preserves matrix multiplication, the kernel is trivial.
Finally, the First-Isomorphism Theorem gives the result. □

Definition 0.1.3. Let F be a field. Then, a set

$$T_n^+(F) \overset{\text{def}}{=} \{A \in \mathcal{M}_{n\times n}(F) \mid A \text{ is upper-triangle matirx with all diagonals } 1\}$$

is called a set of *unit upper triangular matrices*.

Lemma 0.1.4. A unit upper triangular matrices is subgroup of $\mathrm{SL}_n(F)$.

Proof. Closed under the addition can be shown by just calculation. Let $A \in T_n^+(F)$ be given. Then,

$$A = I + (A - I)$$

and the $A - I$ must be nilpotent, because it is strictly upper triangular. Denote $N = A - I$. Now,

$$(I + N) \cdot (I - N + N^2 - N^3 + \cdots + (-1)^{n-1}N^{n-1}) = I + (-1)^{n-1}N^n = I$$

And, $I - N + N^2 - \cdots$ is contained in $T_n^+(F)$, A has an inverse in $T_n^+(F)$. The determinant is clearly 1. □

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0.1.1 General Linear Group over the Finite Field

Proposition 0.1.5. Let $\mathbb{F}_q$ be a finite field of order q. Then,

$$\mathrm{GL}_n(\mathbb{F}_q) \text{ has order } (q^n-1)(q^n-q^1)\cdots(q^n-q^{n-1})$$
$$\mathrm{SL}_n(\mathbb{F}_q) \text{ has order } \frac{1}{q-1}|\mathrm{GL}_n(\mathbb{F}_q)|$$

Proof. Note: Clearly, $|M_{n\times n}(\mathbb{F}_q)| = q^{n^2}$.
To form an invertible matrix, we choose n linearly independent columns.
Choice of the first column as non-zero: there are $q^n - 1$ possibilities.
And, the second column can be any vector except a multiple of the first, then $q^n - q$ possibilities.
Successively, choose the third as any vector but not a linear combination of the first and second, then there are $q^n - q^2$ (More specifically, q^2 is the number of cases of $a_1u_1 + a_2u_2$).
Inductively, we obtain that

$$|\mathrm{GL}_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1}(q^n - q^k)$$

And order of $\mathrm{SL}_n(\mathbb{F}_q)$ is obtained by the above lemma. □

Note: Since the properties of the finite field, the order q must be a power of a prime.